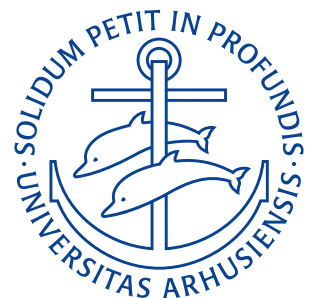


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Lecture Notes

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Contents

1	Lecture 1: Introduction	2
2	Introduction	2
2.1	Coordinate Rotation	2
2.2	Cartesian Tensors	2
2.2.1	Algebra of Cartesian Tensors	3
2.3	Operational Tensors	3
3	Lecture 2: Traction, Stress, and Equilibrium	4
4	Traction, Stress, and Equilibrium	4
4.1	State of Stress	4
4.2	Components of Stress	4
4.3	Stress at a Point	5
4.3.1	Stress on a Normal Plane	5
4.3.2	Dyadic Representation of Stress	6

Lecture 1: Introduction

26. Januar 2026

2 Introduction

This collection of notes are the personal notes of me, Noah Hansen, for the course “Theory of Elasticity” (Da: Elasticitetsteori) taught on the 4th semester of the bachelor on the mechanical engineering programme at Aarhus Universitet.

The notes will take base in the lectures by Assistant Professor Matteo Pezzulla and “Introduction to Linear Elasticity” by Phillip L. Gould and Yuan Feng.

2.1 Coordinate Rotation

Given a position vector $\mathbf{r}$ to a point P , we can resolve it into components with respect to two different Cartesian systems, x_i and x'_i , having a common origin.

The point P has coordinates $P(x_1, x_2, x_3) = P(x_i)$ in the unprimed system and $P(x'_1, x'_2, x'_3) = P(x'_i)$ in the primed system. The linear transformation between the coordinates is then:

$$\begin{aligned}x'_1 &= \alpha_{11}x_1 + \alpha_{12}x_2 + \alpha_{13}x_3 \\x'_2 &= \alpha_{21}x_1 + \alpha_{22}x_2 + \alpha_{23}x_3 \\x'_3 &= \alpha_{31}x_1 + \alpha_{32}x_2 + \alpha_{33}x_3\end{aligned}$$

or

$$x'_i = \alpha_{ij}x_j.$$

Each of the nine quantities α_{ij} is the cosine of the angle between the i th primed and the j th unprimed axis, i.e.

$$\alpha_{ij} = \cos(x'_i, x_j) = \frac{\partial x_j}{\partial x'_i} = \mathbf{e}'_i \cdot \mathbf{e}_j = \cos(\mathbf{e}'_i, \mathbf{e}_j).$$

These α_{ij} 's are known as direction cosines.

2.2 Cartesian Tensors

A tensor of order n is a set of 3^n quantities which transform from one coordinate system x_i to another x'_i , by the following law:

Table 1: Transformation laws for Tensors

n	Order	Transformation law
0	Zero (scalar)	$A'_i = A_i$
1	One (vector)	$A'_i = \alpha_{ij}A_j$
2	Two (dyadic)	$A'_{ij} = \alpha_{ik}\alpha_{jl}A_{kl}$
3	Three	$A'_{ijk} = \alpha_{il}\alpha_{jm}\alpha_{kn}A_{lmn}$
4	Four	$A'_{ijkl} = \alpha_{im}\alpha_{jn}\alpha_{kp}\alpha_{lq}A_{mnpq}$

Order zero and order one tensors (scalars and vectors are already familiar quantities, whereas higher-order tensors are not. These tensors are, however, useful to describe physical quantities with a corresponding number of associated directions. Second-order tensors (dyadics) are especially prevalent for elasticity and the corresponding transformation may be carried out in matrix format as:

$$\mathbf{A}' = \mathbf{R}\mathbf{A}\mathbf{R}^T.$$

in which

$$\mathbf{R} = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{bmatrix} \quad \text{and} \quad \mathbf{A}' = \begin{bmatrix} A'_{11} & A'_{12} & A'_{13} \\ A'_{21} & A'_{22} & A'_{23} \\ A'_{31} & A'_{32} & A'_{33} \end{bmatrix}.$$

2.2.1 Algebra of Cartesian Tensors

Arithmetic and algebra for tensors is similar to matrix operations when it comes to addition, subtraction, equality, and scalar multiplication. Multiplication of two tensors of order n and m produces a new tensor of order $n + m$, e.g.

$$A_i B_{jk} = C_{ijk}.$$

2.3 Operational Tensors

Additional tensor operations are made possible by the Kronecker delta δ_{ij} , defined as,

$$\delta_{ij} \equiv \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

and the permutation tensor ϵ_{ijk} defined such that:

$$\begin{aligned} \epsilon_{ijk} &= \frac{1}{2}(i-j)(j-k)(k-i) \\ &= \begin{cases} 0 & \text{If any two of } i, j, k \text{ are equal} \\ 1 & \text{For an even permutation (forward on the number line } 1, 2, 3, 1, 2, 3, \dots) \\ -1 & \text{For an odd permutation (backward on the number line } 3, 2, 1, 3, 2, 1, \dots). \end{cases} \end{aligned}$$

The Kronecker delta δ_{ij} changes the subscripts in a tensor by multiplication:

$$\begin{aligned} \delta_{ij} A_i &= \delta_{1j} A_1 + \delta_{2j} A_2 + \delta_{3j} A_3 = A_j \\ \delta_{ij} C_{ijk} &= \delta_{11} C_{11k} + \delta_{22} C_{22k} + \delta_{33} C_{33k} = C_{iik} = A_k. \end{aligned}$$

It is also useful in vector algebra and for coordinate transformations. E.g. starting with the dot product of two unit vectors:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} \quad (1)$$

and seeking the component of vector $\mathbf{A}$ in the j -direction A_j we do:

$$\begin{aligned} \mathbf{e}_j \cdot \mathbf{A} &= (\mathbf{e}_j \cdot \mathbf{e}_i) A_i \\ &= \delta_{ji} A_i \\ &= A_j. \end{aligned}$$

The permutation tensor ϵ_{ijk} is useful for vector cross-product operations:

$$\epsilon_{ijk} A_j B_k \mathbf{e}_i = \mathbf{A} \times \mathbf{B}.$$

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

4 Traction, Stress, and Equilibrium

4.1 State of Stress

For a deformable body subject to external loading we focus on an element with area ΔA_n oriented according to the unit normal $\mathbf{n}$, we call the resultant force $\Delta \mathbf{F}_n$ and the resultant moment $\Delta \mathbf{M}_n$. Both are vector quantities, and are *not*, in general, parallel to $\mathbf{n}$. We represent the intensity of these as

$$\lim_{\Delta A_n \rightarrow 0} \frac{\Delta \mathbf{F}_n}{\Delta A_n} = \frac{d\mathbf{F}_n}{dA_n} = \mathbf{T}_n$$

$$\lim_{\Delta A_n \rightarrow 0} \frac{\Delta \mathbf{M}_n}{\Delta A_n} = \frac{d\mathbf{M}_n}{dA_n} = \mathbf{C}_n$$

where $\mathbf{T}_n$ is known as the stress vector or the *traction*, and $\mathbf{C}_n$ is called the *couple-stress vector*.

4.2 Components of Stress

We consider an infinitesimal rectangular prism erected at the point from [Section 4.1](#) and erect a set of Cartesian coordinates x_i parallel to the sides as shown on [Figure 4.1](#). Each of these coordinate axis has a corresponding unit vector $\mathbf{e}_i$.

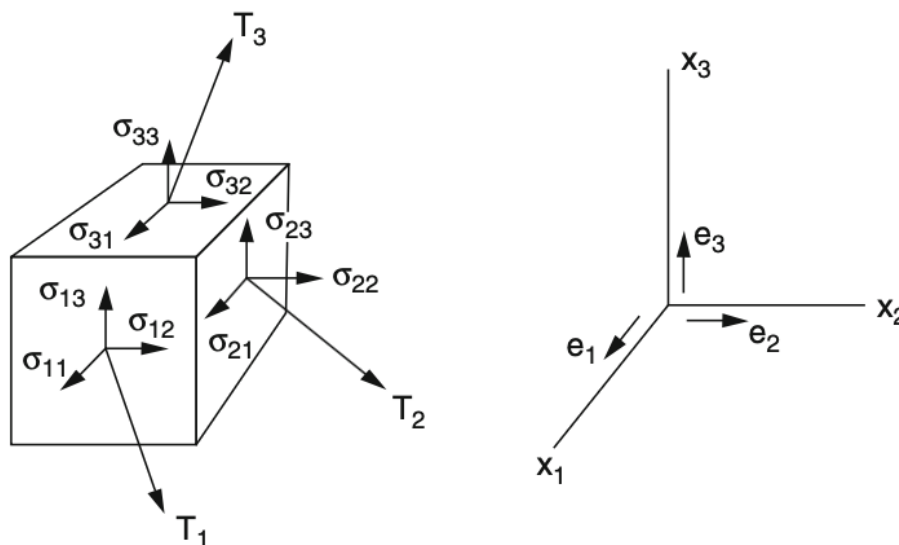


Figure 4.1: Components of stress.

Each traction may be written in terms of its Cartesian components in the form

$$\mathbf{T}_i = \sigma_{ij} \mathbf{e}_j \quad (2)$$

which is shown for $\mathbf{T}_1$ below

$$\mathbf{T}_1 = \sigma_{11} \mathbf{e}_1 + \sigma_{12} \mathbf{e}_2 + \sigma_{13} \mathbf{e}_3.$$

In [Equation \(2\)](#), the coefficients σ_{ij} are known as stresses, and the entire array of stresses forms the so-called Cauchy stress tensor. The subscript and sign convention for stress components σ_{ij} are:

1. The first subscript i refers to the *normal* $\mathbf{e}_i$, which denotes the face on which $\mathbf{T}_i$ acts.
2. The second subscript j corresponds to the *direction* $\mathbf{e}_j$ in which the stress acts.

3. The normal components σ_{ii} are positive if they produce tension and negative if they produce compression. The shearing components $\sigma_{ij} (i \neq j)$ are positive if directed in the positive x_j direction while acting on the face with the unit normal e_i or if directed in the negative x_j direction while acting on the face with normal $-e_i$

4.3 Stress at a Point

It can be shown that to completely describe the state of stress at a point, it is sufficient to specify the tractions T_i on each of the three planes e_i – i.e. this is equivalent to specifying the nine stress components σ_{ij} .

$$T_i = \sigma_{ji} n_j. \quad (3)$$

And if the components T_i are known the magnitude of T_n is evaluated as

$$T_n = |T_n| = (T_i T_i)^{\frac{1}{2}}.$$

4.3.1 Stress on a Normal Plane

It is sometimes useful to resolve T_n into components that are normal and tangential to the differential surface element dA_n as depicted on Figure 4.2. Here, the normal component σ_{nn} is calculated by

$$\begin{aligned} \sigma_{nn} &= |N| = T_n \cdot n \\ &= T_i e_i \cdot n = T_i n_i. \end{aligned}$$

or using Equation (3) as

$$\sigma_{nn} = \sigma_{ji} n_j n_i. \quad (4)$$

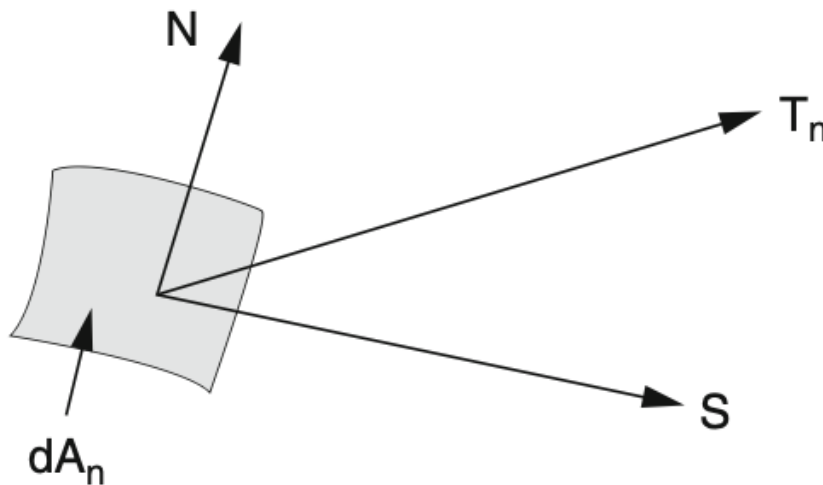


Figure 4.2: Differential surface element.

The tangential component σ_{ns} is

$$\begin{aligned} \sigma_{ns} &= |S| = T_n \cdot s \\ &= T_i e_i \cdot s \\ &= T_i s_i \\ &= \sigma_{ji} n_j s_i. \end{aligned}$$

where

$$s_i = \mathbf{e}_i \mathbf{s}.$$

Often it is easy to calculate σ_{ns} using the Pythagorean theorem as

$$\sigma_{ns} = \sqrt{T_i T_i - \sigma_{nn}^2}.$$

One is also able to resolve N and S into Cartesian components using (k) to designate the axes as

$$\begin{aligned}\sigma_{nn(k)} &= \mathbf{N} \cdot \mathbf{e}_k = \sigma_{nn} \mathbf{n} \cdot \mathbf{e}_k \\ &= \sigma_{nn} n_k \\ &= \sigma_{ji} n_j n_i n_k, \quad k = 1, 2, 3.\end{aligned}$$

For σ_{ns} subtraction gives

$$\sigma_{ns(k)} = T_k - \sigma_{nn(k)} \quad k = 1, 2, 3$$

which may also be written as

$$\sigma_{ns(k)} = \sigma_{jk} n_j - \sigma_{ji} n_j n_i n_k = (\delta_{ki} - n_i n_k) \sigma_{ji} n_j.$$

4.3.2 Dyadic Representation of Stress

One is able to view the stress tensor as a vector-like quantity with a magnitude and associated direction(s) specified by unit vectors. The *dyadic* is such a representation. We write the stress tensor or stress dyadic as

$$\begin{aligned}\boldsymbol{\sigma} &= \sigma_{ij} \mathbf{e}_i \mathbf{e}_j \\ &= \sigma_{11} \mathbf{e}_1 \mathbf{e}_1 + \sigma_{12} \mathbf{e}_1 \mathbf{e}_2 + \sigma_{13} \mathbf{e}_1 \mathbf{e}_3 + \sigma_{21} \mathbf{e}_2 \mathbf{e}_1 + \sigma_{22} \mathbf{e}_2 \mathbf{e}_2 + \sigma_{23} \mathbf{e}_2 \mathbf{e}_3 + \sigma_{31} \mathbf{e}_3 \mathbf{e}_1 + \sigma_{32} \mathbf{e}_3 \mathbf{e}_2 + \sigma_{33} \mathbf{e}_3 \mathbf{e}_3\end{aligned}$$

wherein the double vectors are called *dyads*. The corresponding tractions are evaluated by an operation analogous to the dot product:

$$\mathbf{T}_i = \boldsymbol{\sigma} \cdot \mathbf{e}_i = \sigma_{ij} \mathbf{e}_j.$$

Similarly, the normal and tangential components of the traction $\mathbf{T}_n$ on a plane defined by normal $\mathbf{n}$ are

$$\begin{aligned}\sigma_{nn} &= \boldsymbol{\sigma} \cdot \mathbf{n} \cdot \mathbf{n} \\ &= \mathbf{T}_n \cdot \mathbf{n} \\ &= \sigma_{ij} n_i n_j\end{aligned}$$

and

$$\begin{aligned}\sigma_{ns} &= \boldsymbol{\sigma} \cdot \mathbf{n} \cdot \mathbf{s} \\ &= \mathbf{T}_n \cdot \mathbf{s} \\ &= \sigma_{ij} n_i s_j.\end{aligned}$$